The Roadmap of Audio Interaction

Stage 1

Speech Components

Discrete Speech Representation

- HuBERT
- CosyVoice 2
- SoundStream
- EnCodec
- WavTokenizer
- SpeechTokenizer
- X-Codec
- Mimi

Continuous Speech Representation

- tGSLM
- Spectron
- LauraGPT
- Mini-Omni
- SLAM-Omni
- Flow-Omni
- CALM
- HoliTok

Stage 2

Half-Duplex Speech Interaction

Text-Centric Cascaded Interaction

- POMDP SDS
- SimpleTOD
- AudioGPT
- ESPnet-SDS
- Eva
- FireRedChat
- LTS-VoiceAgent
- DDTSR

Thinker-Talker Cascaded Interaction

- LLaMA-Omni
- Freeze-Omni
- MinMo
- Uni-MoE
- Qwen2.5-Omni
- LLaMA-Omni 2
- SALMONN-omni
- Qwen3-Omni
- Uni-MoE-2.0-Omni
- Qwen3.5-Omni

Native End-to-End Speech Interaction

- SpeechGPT
- USDM
- Mini-Omni
- GLM-4-Voice
- Baichuan-Audio
- Kimi-Audio
- Step-Audio 2
- MOSS-Speech

Stage 3

Full-Duplex Speech Interaction

Module-Level External Control

- TurnGPT
- VAP
- Easy Turn
- FlexDuo
- FireRedChat
- SoulX-Duplug
- SID
- X-Talk

Semi-Embedded Semantic Assistance

- Freeze-Omni
- MinMo
- BC head
- VITA

Native Dynamic Interaction

- dGSLM
- LSLM
- Moshi
- SyncLLM
- OmniFlatten
- NTPP
- F-Actor
- SALMONN-omni
- MiniCPM-o 4.5
- D2

Stage 4

Agentic Audio Interaction

Interaction with Tool Environment

- Stream RAG
- Heard Enough to Act
- SIA
- Qwen-Audio-Agent
- PACE
- DuplexAgent-RSI

Interaction with Agents

- SpeechAgents
- SALMONN-duo
- Gander
- Qwen-Audio-Agent
- AdaptDuplex
- Jarvis

Interaction with Robotic Embodiment

- VLAS
- RoboOmni
- Interactive Language
- YAY Robot
- VITA-E
- MIRA

Interaction with Self-Memory & Lifelong Context

- EgoMem
- VoiceMem
- BELIEFMEM
- Voice Memory for Agentic ASR
- VoxMem
- MemEval

Expansion of Interaction Scope

Representation Foundations

How is speech represented?

From Text-Mediated to Speech-Native Interaction

How does speech participate in language?

From External Control to Native Dynamic Coordination

How are speaking and listening coordinated?

Beyond Dialogue: Action, Feedback and Collaboration

How does speech guide actions and collaboration?